Supplementary Information for:Helium Step Heats on Geothite at Water Saturated Conditions Show Diffusive Helium Loss and High Retentivity at Surface Temperatures

Ben J. Thyer[a,*], Jonathan Treffkorn[a], Kenneth A. Farley[a]

[a]*Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, 91125, CA, USA*

S1. Permeation Lag Correction and Ampoule Permeability

S1.1. Permeation Lag Correction

Because HT-Win-06-01b was encapsulated in an imperfectly helium-transmissive silica glass ampoule, permeation lag was an expected complication in these measurements. To address this, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C) steps where the helium in goethite at high pH2O was effectively immobile at experimental timescales, following the permeation step method of Thyer et al. (2026). For 3He, measured signals at these low temperature permeation steps were entirely controlled by the process of permeation through silica glass, and therefore could be used to estimate the amount of helium remaining within the ampoule between high-temperature experimental steps. However, for 4He, the measured signals at low temperature permeation steps were most often indistinguishable from the blank, so Eq. (4) of Thyer et al. (2026) could not be applied to 4He and the permeation step method could not be carried through for that isotope.

As a result, permeation lag is instead addressed using the bounding method described in Thyer et al. (2026), which constrains the ampoule's residual helium from above rather than measuring it. In this approach, an

*Corresponding author.
Email address: benjthyer@gmail.com (Ben J. Thyer)

upper bound constraint for the helium content of the ampoule at the end of each high-temperature step was calculated using Eq. (7) of Thyer et al. (2026). 3He was reduced this way as well, for consistency with 4He; because 3He can also be reduced by the permeation step method, the two reductions can be compared directly; that comparison, an internal test of the bounding method, is given by Thyer (2026).

Both routes require the ampoule's permeation rate constant λ, which was measured during the experiment itself rather than assumed from the literature (Thyer et al. 2026). This in-experiment measurement is not an alternative method of handling permeation lag but a prerequisite of the permeation step method and, here, the empirical basis for choosing a conservative λ for the bounding method. It is indispensable in this experiment: the permeability of ampoules used in water vapor saturated step heats cannot reasonably be assumed to match literature values for anhydrous silica glass.

Ten series of 3–4 consecutive permeation steps held at a single temperature between 90–110°C for 120-180 minutes each with 26 minutes of cooling time before being inlet into the Helix-MC-Plus^{10K} were identified as suitable for this exercise. The 3He measurements in these series were fit to Eq. (6) of Thyer et al. (2026) using a non-linear least squares optimization (Fig. S1). This is accomplished with the Levenberg-Marquardt algorithm implemented in the curve_fit function in the SciPy package of Python (Virtanen et al., 2020), weighted by each measurement's own uncertainty. One additional series that otherwise qualified was not well fit by this model and was not used further for clarity, though its fit value of λ was within uncertainty of the other series' fit values for λ.

An initial guess of $\lambda \approx 6{\times}10^{-6}$ sec^{-1} and $n_0 \approx \frac{m_i}{1-e^{-\lambda t_i}}$ (evaluated at the first measurement of a series) pccSTP was passed to the optimizer for each series. Uncertainties from the model fits were calculated using the covariance matrices generated for each of the model fits. Although the initial helium amount n_0 was a free parameter in the optimizations, the rate constant of permeation (λ) was the only parameter of interest for this exercise. The fitted rate constants span 2.7×10^{-5} to 6.3×10^{-5} s^{-1}, with 2σ uncertainties of 0.08 to 0.46 log units and no trend resolvable against step duration:

All measured apparent 3He permeabilities were equal to or in excess of the literature values for anhydrous silica glass (Fig. S2), similar to observations discussed in Thyer et al. (2026). 4He permeability itself could not be independently measured in this experiment due to blank-level signals during the low temperature permeation steps and is assumed to similarly exceed the

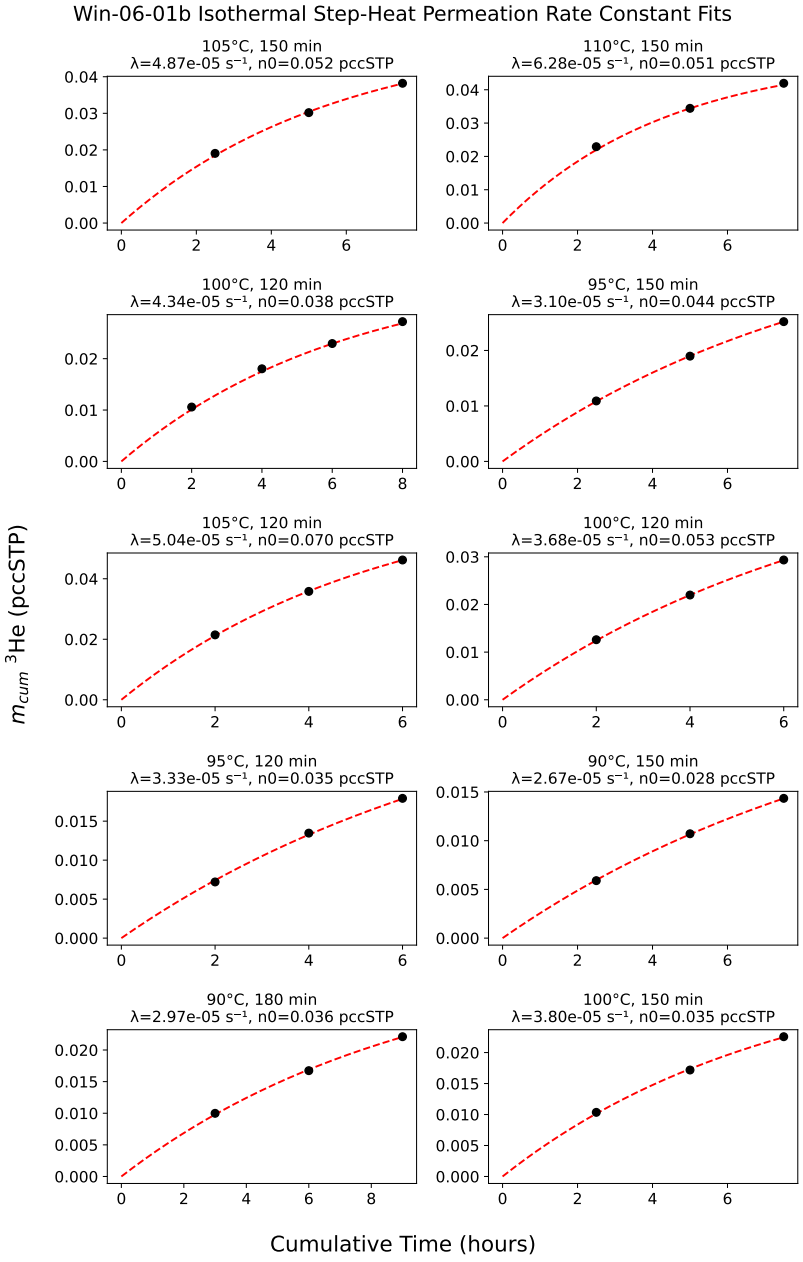


Figure S1: Cumulative yield against cumulative time for helium-3 yields for isothermal permeation steps. A non-linear least squares optimization fits λ and n_0 of Eq. (6) of Thyer et al. (2026) to data from a series of permeation steps like these. The permeability K is calculated according to the relation $K = \lambda/GRT$ and is plotted as a function of temperature in Fig. S2. The curvature of each series is what conditions the fit: a series in which the permeation rate has not yet slowed appreciably constrains λ and n_0 only weakly.

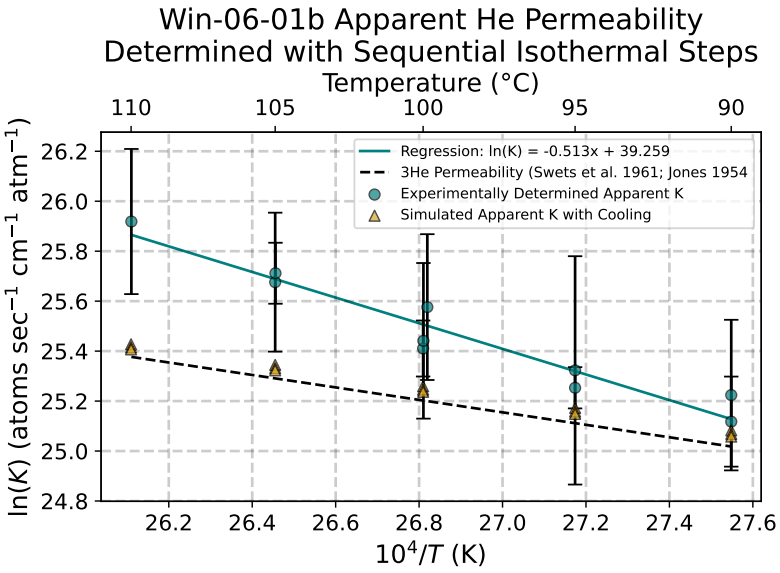


Figure S2: Arrhenius plot of apparent permeation coefficients as a function of inverse temperature for HT-Win-06-01b, converted from the rate constants fitted in Fig. S1 via $K = \lambda/GRT$. The activation energy is higher than anhydrous silica glass, as expected (Masaryk, 1968; Shelby, 1972). It is unexpected that the glass ampoule of HT-Win-06-01b has higher apparent 3He permeability than the literature values of Swets et al. (1961) and Jones (1953), but this can be explained by an increase in the partial pressure of the gas phase within the tube due to displacement by high He diffusivity, low He solubility water. Yellow points labeled "Diffusion Model with cooling, Anhydrous" were produced to evaluate the expected excess apparent permeability due to the 25 minute cooldown period before measurement. This is accomplished by simulating the measured apparent permeability for a dry silica glass ampoule that continued to permeate helium while cooling for an additional 26 minutes before measurement after being held at 90–110°C for 120–180 minutes, similarly to the measured values. Variability in these points is due to differences in the length of time the ampoule was heated but is very small. They show a near constant offset from the expected values of Swets et al. (1961) and Jones (1953) that is much smaller than the uncertainty in the regression of the fit values determined experimentally in teal.

literature values for anhydrous silica glass. To evaluate the uncertainty due to the permeation lag, we use the literature values as a conservative (lower bound) estimate of the true permeability; a λ that underestimates the ampoule's true rate constant yields a conservative (higher) upper bound on the residual helium (Thyer et al. 2026). λt (where $\lambda = GRTK$) is estimated here using the 4He permeability K reported by Swets et al. (1961), while the 3He permeability is taken to be the product of the 4He permeability and the relative 3He/4He permeability reported by Jones (1953). The geometric constant G is calculated using the measured inner and outer diameter of the ampoule (Fig. S3).

The first three experimental steps had values of λt lower than 4, and therefore the upper bound of the helium content of these steps could not be determined. These measurements are omitted from further analysis. Later experimental steps with values of λt lower than 4 could be determined using Eq. (S18) from the supplement to Thyer et al. (2026) because upper bounds on previous steps' residual helium content could be evaluated.

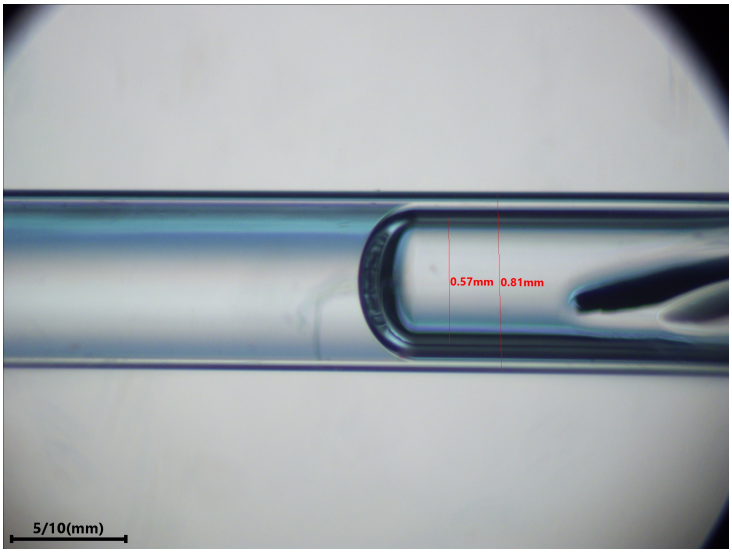


Figure S3: Measurements of the inner and outer diameter of the ampoule used in the water vapor saturated step heats of Win-06-01b. The inner walls of the ampoule are only clearly visible within air bubbles.



Figure S4: A view of the entire experimental ampoule after the completion of the step heat experiment before it was broken for the final step in the resistance furnace. At room temperature, the tube is approximately half-filled with gas, one-third filled with liquid water, and one-sixth filled with <2 mm long shards of irradiated Win-06-01b goethite. A gel-like substance was observed lining the inner walls of the tube, which is likely to be amorphous silica dissolved from the tube and redeposited by the thermally-cycled, high pressure liquid water phase.

S2. Analytical Methods

S2.1. Final Step

A hot furnace blank subjected to the same volume splits as the final HT-Win-06-01b measurement gave a 3He blank of 6.8 fccSTP and a 4He blank of 0.31 nccSTP. This blank correction was less than 1% of the measured 4He and negligible for 3He. A re-extract of HT-Win-06-01b immediately following the measurement yielded above-blank 3He and 4He signals, which were added to the total measurement, although both constituted less than 1% of the final step yield. Sensitivity was calculated by measuring a 3He-enriched air standard with known 3He and 4He contents immediately before the measurement.

S3. Kinetic mechanisms and the limits of a single heating schedule

Solid-state rate laws separate the temperature dependence of the rate from its dependence on the extent of reaction (Vyazovkin et al., 2011):

$$\frac{d\alpha}{dt} = A \exp\left(-\frac{E_a}{RT}\right) h(\alpha), \tag{S1}$$

where α is the fraction of goethite converted. The reaction model $h(\alpha)$ expresses the mechanism: it describes how the rate changes as conversion proceeds, which depends on where and how the reaction advances through a grain. For first-order kinetics (F1), grains convert independently once nucleated, so the rate is proportional to the goethite remaining, $h = 1 - \alpha$. For a reaction front moving inward from the grain surface (R2, R3), the rate is proportional to the shrinking area of the unreacted core. For random nucleation followed by growth (A2, A3), the rate first accelerates and then slows. For any heating schedule, each rate law gives α as a function of a single variable,

$$x = A \sum_j \Delta t_j \exp\left(-\frac{E_a}{RT_j}\right), \tag{S2}$$

the Arrhenius-weighted time delivered by the steps up to the current one; for a single isothermal step, $x = kt$. We tested the mechanisms in Table S1, including spherical diffusion on the same footing: for diffusion, x is the Fourier number Dt/a^2 with D/a^2 Arrhenius in T. Under vacuum, the driving-force factor $1 - \exp(-\Delta\mu/RT)$ is unity to better than 10^{-9}, so x depends only on the heating schedule. Diffusion and conversion then differ only in the shape of $\alpha(x)$ and in the fitted E_a.

Table S1: Kinetic models: reaction model $h(\alpha)$ and conversion as a function of the Arrhenius-weighted time, $\alpha(x)$.

Model	$h(\alpha)$	$\alpha(x)$
R2 contracting cylinder	$2(1-\alpha)^{1/2}$	$1-(1-x)^2$
R3 contracting sphere	$3(1-\alpha)^{2/3}$	$1-(1-x)^3$
F1 first order	$1-\alpha$	$1-e^{-x}$
A2, A3 Avrami ($n=2,3$)	$n(1-\alpha)[-\ln(1-\alpha)]^{1-1/n}$	$1-e^{-x^n}$
D3 sphere diffusion	—	$1-\frac{6}{\pi^2}\sum_{m=1}^{\infty}\frac{e^{-m^2\pi^2 x}}{m^2}$

S3.1. Fits to the vacuum data

Table S2 gives the fit of each model to the cumulative KAF ^{3}He release within the fitting window. It also gives the misfit of the resulting out-of-sample prediction of HDM, and the fit to the KAF water data. F1 gives the best fit and the best prediction. R3 and R2 are close behind, so we describe the mechanism as nucleation- or interface-controlled rather than claim first order specifically. The Avrami fits reach the lower bound of E_a and fail on both data sets. Spherical diffusion with $\phi = 0$ fits KAF 4.4 times worse than F1 and cannot produce the $F \approx 0.89$ plateau. To compare diffusion on the same footing, we also fit D3 with ϕ fixed at the F1 value and with ϕ free, over the same 18 KAF steps. The imposed plateau improves the KAF misfit to 0.026 and 0.018, which still ranks below F1, R3 and R2, and the out-of-sample prediction of HDM does not improve (rms 0.057 and 0.055, against 0.018 for F1). These fits also require E_a of 242–287 kJ mol^{-1} and a rate factor $s = 0.73$ for the trap-isolated steps, although a diffusive process should not respond to the water pressure at all.

The fitting window excludes the low-temperature steps, where release is dominated by diffusion from goethite, and the plateau, where it is dominated by diffusion from hematite; the rms misfits are computed over the same steps. Repeating the fits with every step included gives the same result: F1 remains the best model ($E_a = 122.7$ kJ mol^{-1}, $\phi = 0.112$, rms 0.0073 for KAF and 0.026 for the HDM prediction), followed by R3 (99.4 kJ mol^{-1}; 0.0097, 0.030) and R2 (87.9 kJ mol^{-1}; 0.012, 0.033), with D3 at 175.9 kJ mol^{-1} (0.063, 0.061; 0.021 and 0.043 with ϕ fixed at the F1 value).

Table S2: Kinetic and Diffusive Model fits to the vacuum experiments (from Monteiro (2017) and Farley et al. (2024)). s: rate factor for the trap-isolated steps. All rms values are over the same 18 KAF and 19 HDM steps. D3 is fitted with ϕ fixed at 0, fixed at the F1 value, and free; the water fit does not depend on ϕ and is given once.

Model	E_a	$\ln A$	ϕ	s	rms KAF	rms HDM	E_a (water)
R2	92.8	10.63	0.149	0.50	0.0133	0.0291	66.1
R3	103.1	12.73	0.136	0.49	0.0114	0.0240	74.1
F1	120.8	18.15	0.101	0.47	0.0087	0.0176	90.9
A2	25.5	−4.73	0.000	0.20	0.0212	0.1906	20.0[a]
A3	20.0[a]	−5.85	0.203	0.00	0.0433	0.2342	20.0[a]
D3, $\phi = 0$	196.5	32.60	0.000	1.33	0.0384	0.0478	184.6
D3, $\phi = \phi_{\mathrm{F1}}$	241.5	43.66	0.101	0.73	0.0257	0.0567	—
D3, ϕ free	286.8	54.72	0.185	0.48	0.0176	0.0549	—

E_a in kJ mol^{-1}, A in s^{-1}. [a]At the lower bound of the fit.

S4. Thermodynamics of Goethite Dehydroxylation

Goethite breaks down to hematite by the dehydroxylation reaction

$$2\,\alpha\text{-FeOOH} \;\rightleftharpoons\; \alpha\text{-Fe}_2\text{O}_3 + \text{H}_2\text{O}_{(\text{fluid})}, \tag{S3}$$

whose Gibbs energy change, written per mole of hematite formed, is

$$\Delta G_r(T,P) = G_{\mathrm{hem}}(T,P) + G_{\mathrm{w}}(T,P) - 2\,G_{\mathrm{goe}}(T,P) + \Delta G_{\mathrm{surf}}. \tag{S4}$$

where ΔG_r is the Gibbs energy change of the reaction and G_{hem}, G_{w}, and G_{goe} are the Gibbs energies of hematite, water, and goethite, respectively. ΔG_{surf} is the surface energy contribution, which must be considered to reproduce observed phase stability (Navrotsky et al., 2008). Goethite is stable where $\Delta G_r > 0$ and hematite plus water where $\Delta G_r < 0$. The values and equations used to evaluate each of these terms are given below.

Gibbs energies of the solids. The Gibbs energy of a phase at (T, P) follows from integrating $dG = -S\,dT + V\,dP$ from the reference state $(T_r, P_r) = (298.15\ \mathrm{K}, 1\ \mathrm{bar})$, first in temperature at P_r and then in pressure at T. At P_r, $H(T) = H(T_r) + \int_{T_r}^{T} C_p\,dT$ and $S(T) = S(T_r) + \int_{T_r}^{T} (C_p/T)\,dT$, so $G(T,P_r) = H(T) - TS(T)$. Adding $\int_{P_r}^{P} V\,dP$, and treating the solids as incompressible with no thermal expansion, gives

$$G_i(T,P) = \Delta_f H_i^0 + \int_{T_r}^{T} C_{p,i}\,dT \;-\; T\left[S_i^0 + \int_{T_r}^{T} \frac{C_{p,i}}{T}\,dT\right] \;+\; V_i\,(P - P_r), \tag{S5}$$

where $H(T_r) = \Delta_f H_i^0$ and $S(T_r) = S_i^0$. This is the "apparent" Gibbs energy of formation (Berman, 1988); the element-entropy terms that distinguish it from the true Gibbs energy of formation cancel in any balanced reaction. Compressibility and thermal expansion change V_i by $\lesssim 1\%$ over the conditions considered and are neglected.

Goethite heat capacity uses the Berman–Brown form (Berman and Brown, 1985) refined by Majzlan et al. (2003a), $C_p = k_0 + k_1 T^{-1/2} + k_2 T^{-2} + k_3 T^{-3}$. Hematite uses the same polynomial with the coefficients of Berman (1988), which Majzlan et al. (2003a) held fixed in their refinement. Both heat capacities are integrated analytically.

Water. Water is evaluated with the IAPWS-95 equation of state (Wagner and Pruß, 2002), implemented in CoolProp (Bell et al., 2014), which is valid for liquid, vapor and supercritical water to 1 GPa. In the same apparent convention,

$$G_{\mathrm{w}}(T,P) = \Delta_f H_{\mathrm{w}}^0 + [h(T,P) - h(T_r,P_r)] - T\left[S_{\mathrm{w}}^0 + s(T,P) - s(T_r,P_r)\right], \tag{S6}$$

where h and s are the IAPWS-95 molar enthalpy and entropy and the reference state is liquid water at T_r, P_r. Table S3 lists all input data. We use the enthalpies and entropies of Majzlan et al. (2003a) refined by mathematical programming against the phase-equilibrium brackets of Voigt and Will (1981).

As a check, at 298.15 K and 1 bar with liquid water, Eqs. (S4)–(S6) give $\Delta G_r = -0.26$ kJ: hematite plus liquid water is favored by only 0.13 kJ per mole of FeOOH. This is well inside the ± 1.4 kJ mol^{-1} uncertainty that Majzlan et al. (2003a) assign to the goethite–hematite free-energy difference; the two assemblages are essentially balanced at ambient conditions. We carry this uncertainty (± 2.8 kJ per reaction as written) as bounding curves on the phase diagram.

Surface energy. A crystal of finite size carries an excess free energy at its surface in addition to the free energy of its bulk (Gibbs, 1878). For a population of particles of fixed size, this excess is γA, where γ is the surface energy (J m^{-2}) and A the total surface area. Both the bulk and surface terms scale with the amount of material at fixed particle size, so the surface contribution can be written per mole, $G = G_{\mathrm{bulk}} + \gamma A_m$, with A_m the surface area per mole. A_m grows as $1/d$, so the term is negligible for coarse crystals and significant for nanoparticles (Navrotsky et al., 2008). For the reaction

Table S3: Thermodynamic data used for Eq. (S4).

	Goethite (α-FeOOH)	Hematite (α-Fe_2O_3)
$\Delta_f H^0$ ($\mathrm{kJ\,mol^{-1}}$)	-561.4	-825.8
S^0 ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	59.5	87.4
V^0 ($\mathrm{cm^3\,mol^{-1}}$)	20.881	30.274
C_p coefficients	$k_0 = 133.8765$	$k_0 = 146.86$
	$k_1 = -738.5577$	$k_1 = 0$
	$k_2 = -2.177657 \times 10^6$	$k_2 = -5.5768 \times 10^6$
	$k_3 = 2.165124 \times 10^8$	$k_3 = 5.2563 \times 10^8$
C_p(298 K) ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	74.8	104.0
$\gamma_{\mathrm{hydrated}}$ ($\mathrm{J\,m^{-2}}$)	0.60 ± 0.10	0.75 ± 0.16
$\gamma_{\mathrm{anhydrous}}$ ($\mathrm{J\,m^{-2}}$)	0.91 ± 0.09	1.90 ± 0.30

Enthalpy and entropy: Majzlan et al. (2003a); goethite C_p: Majzlan et al. (2003b,a); hematite C_p: Berman (1988); surface energies: Navrotsky et al. (2008). Units: k_0 $\mathrm{J\,mol^{-1}\,K^{-1}}$, k_1 $\mathrm{J\,mol^{-1}\,K^{-1/2}}$, k_2 $\mathrm{J\,K\,mol^{-1}}$, k_3 $\mathrm{J\,K^2\,mol^{-1}}$; l_1 $(\mathrm{J\,mol^{-1}})^{1/2}\,\mathrm{K^{-1}}$, l_2 $(\mathrm{J\,mol^{-1}})^{1/2}\,\mathrm{K^{-2}}$. Water: $\Delta_f H^0_{\mathrm{w}} = -285.83\ \mathrm{kJ\,mol^{-1}}$, $S^0_{\mathrm{w}} = 69.95\ \mathrm{J\,mol^{-1}\,K^{-1}}$; IAPWS-95 for $h(T,P)$, $s(T,P)$.

as written (one mole of hematite from two moles of goethite),

$$\Delta G_{\mathrm{surf}} = \gamma_{\mathrm{hem}} A_{m,\mathrm{hem}} - 2\,\gamma_{\mathrm{goe}} A_{m,\mathrm{goe}}, \tag{S7}$$

where $A_{m,\mathrm{hem}}$ is per mole of Fe_2O_3 and $A_{m,\mathrm{goe}}$ per mole of FeOOH.

A size must be chosen for each phase. We assume pseudomorphic dehydroxylation, in which hematite inherits the external envelope of the parent goethite, so that both phases share the same external surface area per mole of Fe. This is also the basis of the stability diagrams of Navrotsky et al. (2008). For spherical goethite grains of diameter d this shared area is

$$A_{\mathrm{Fe}} = \frac{6\,V_{\mathrm{goe}}}{d} \quad \text{per mole of Fe.} \tag{S8}$$

Goethite has one Fe per formula unit, so $A_{m,\mathrm{goe}} = A_{\mathrm{Fe}}$. Hematite has two, so $A_{m,\mathrm{hem}} = 2A_{\mathrm{Fe}}$. Substituting into Eq. (S7),

$$\Delta G_{\mathrm{surf}} = \gamma_{\mathrm{hem}}\,(2A_{\mathrm{Fe}}) - 2\,\gamma_{\mathrm{goe}}\,A_{\mathrm{Fe}} = 2A_{\mathrm{Fe}}\,(\gamma_{\mathrm{hem}} - \gamma_{\mathrm{goe}})\,. \tag{S9}$$

The factor of two counts the two moles of Fe that pass through the reaction. It multiplies both terms because both phases carry the same area per Fe. For $d = 10$ nm, $A_{\mathrm{Fe}} = 6 \times 20.881 \times 10^{-6}/10^{-8} = 1.253 \times 10^4\ \mathrm{m^2\,mol^{-1}}$, and with hydrated surface energies $\Delta G_{\mathrm{surf}} = 2 \times 1.253 \times 10^4 \times (0.75 - 0.60) =$

+3.76 kJ. The corresponding values are +0.38 kJ at 100 nm and +0.04 kJ at 1 µm. A positive ΔG_{surf} stabilizes goethite. If the two phases are instead compared at equal particle diameter, the same hydrated surface energies slightly destabilize fine goethite (−1.41 kJ at 10 nm). The choice affects only the stability discussion; the kinetic analysis uses bulk driving forces.

We use hydrated surface energies. They are referenced to liquid water, so they apply wherever the surfaces are in equilibrium with water at or near saturation, which includes all of the water vapor saturated experiments. Anhydrous surface energies apply only to surfaces that have lost their chemisorbed water, i.e. under vacuum.

Phase diagram. The phase diagram (Fig. S5) is computed by solving $\Delta G_r(T, P) = 0$ (Eqs. (S4)–(S9)) numerically for $P = P_{\mathrm{H_2O}}$ at each temperature, using Brent's method on $\log_{10} P$ between 10^{-12} and 10^4 bar, with the solids at the fluid pressure. Where no root exists below the melting curve of ice, the curve is truncated.

Driving force and the role of water. The quantity that matters for kinetics is the driving force for hematite formation, $\Delta\mu \equiv -\Delta G_r/N_A$ (per mole of Fe_2O_3). Water controls $\Delta\mu$ through its own chemical potential, because it is a product of reaction (S3). G_{w} in Eq. (S4) is the chemical potential of H_2O at the prevailing water fugacity. It depends on the partial pressure of water, not on the total pressure. An inert gas, or any other mechanical load, acts only through the solid-volume term $\Delta V_s(P - P_r)$, which contributes $-1.15\ \mathrm{J\,bar^{-1}}$ and slightly favors the denser hematite. At 40 bar that is 46 J, three orders of magnitude smaller than the effect of 40 bar of water. The water-buffered step-heating technique therefore stabilizes goethite primarily by reducing the driving force of the reaction. Surface hydration is a second-order correction that matters only for nanometer grains (Eq. (S9)).

S4.1. Stability and driving force

The model can be used to calculate the stability field for goethite with respect to dehydroxylation to hematite and water in $\log P_{\mathrm{H_2O}}$–T space (Fig. S5). This phase diagram communicates several important features of the goethite–hematite–water system that are relevant to the experiments considered here:

1. Goethite is not stable at the conditions of any of the experiments considered here, in water or in vacuum, in spite of the evidence that it did not react in the water-saturated experiment. Thus, we hypothesize that the reaction is kinetically inhibited for goethite under these conditions.

2. The stability of goethite is strongly dependent on the partial pressure of water below and up to water saturation. Above the saturation pressure, water is liquid and contributes far less to the reaction volume than it does as a gas, becoming a secondary control on goethite's stability.
3. The stability of goethite is strongly dependent on grain size, with smaller grains being more stable than larger grains. At water saturation, bulk goethite is stable below 18 °C, 100 nm goethite below 28 °C, and 10 nm goethite below 108 °C.

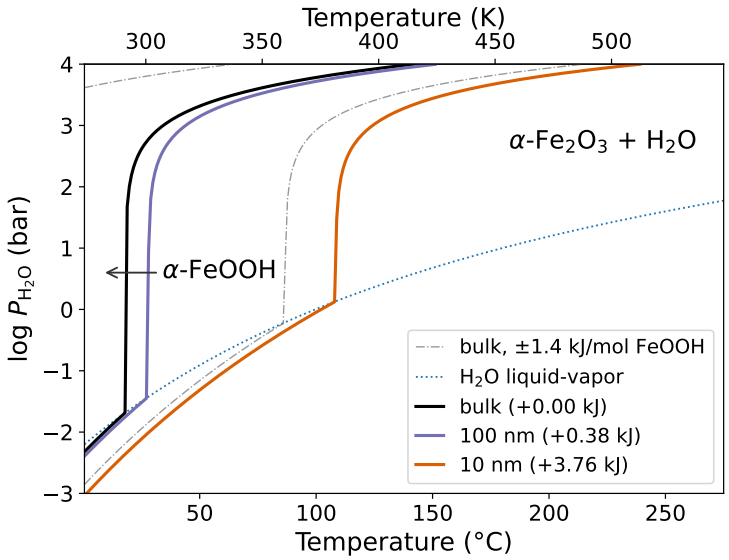


Figure S5: Goethite–hematite–water equilibrium in $\log P_{H_2O}$–T space for bulk goethite and for 100 and 10 nm grains with hydrated surfaces, compared at equal surface area per mole of Fe (Eq. (S9)). Goethite is stable above and to the left of each curve. The dash-dot curves illustrate the absolute uncertainty: it is the bulk stability boundary shifted by the ± 1.4 kJ mol^{-1} FeOOH uncertainty of Majzlan et al. (2003a). The blue dotted curve shows the water vapor saturation pressure; liquid water is present at and above this curve. The thermodynamic data used to compute the phase diagram are in Table S3.

Stability alone does not determine what happens in a step heat; the rate of conversion controls whether this equilibrium state is reached or if the reaction proceeds at all, and that depends on the chemical potential of the reaction: $\Delta\mu$. Table S4 gives the bulk driving forces at the experimental temperatures, evaluated with IAPWS-95 water. At saturation, $\Delta\mu$ is 6–11 kJ, only 1.6–2.4 times RT. Under the vacuum of a step-heating line the water pressure approaches zero, so the chemical potential of water falls

without bound and $\Delta\mu$ is unbounded. The chemical potential of water is therefore the key variable separating the two experimental settings.

Table S4: Bulk driving force for hematite formation, $\Delta\mu = -\Delta G_r$ (kJ per mole Fe_2O_3), at water saturation. In vacuum ($P_{H_2O} \to 0$) the driving force is unbounded.

T (°C)	P_{sat} (bar)	$\Delta\mu_{sat}$	$\Delta\mu_{sat}/RT$
150	4.76	5.68	1.61
175	8.93	6.88	1.84
200	15.55	8.11	2.06
225	25.50	9.37	2.26
250	39.76	10.66	2.45

S5. Nucleation Model of Goethite Dehydroxylation

Nucleation barrier. In classical nucleation theory (Gibbs, 1878; Volmer and Weber, 1926; Becker and Döring, 1935), reviewed by Christian (2002), Kashchiev (2000) and, for mineral systems, De Yoreo and Vekilov (2003) and Lasaga (1998), a spherical hematite nucleus of radius r has free energy

$$\Delta G(r) = -\frac{4}{3}\pi r^3 \frac{\Delta\mu}{V_{hem}} + 4\pi r^2 \gamma, \tag{S10}$$

where $\Delta\mu$ is the driving force per mole of Fe_2O_3 (Table S4), V_{hem} the molar volume and γ the energy of the interface surrounding the nucleus. Nuclei larger than the critical radius $r^* = 2\gamma V_{hem}/\Delta\mu$ grow, and the energetic barrier to forming one is

$$\Delta G^* = \frac{16\pi\,\gamma^3\,V_{hem}^2}{3\,\Delta\mu^2}. \tag{S11}$$

In the first-order limit of §4.1, the rate constant is a nucleation rate of the form derived by Turnbull and Fisher (1949),

$$k = A\exp\!\left(-\frac{E_a}{RT}\right)\exp\!\left(-\frac{\Delta G^*}{k_B T}\right),$$
$$\frac{\Delta G^*}{k_B T} = \frac{C\gamma^3}{T\,\Delta\mu^2}, \qquad C \equiv \frac{16\pi V_{hem}^2}{3k_B} = 1.112\times 10^{15}\ \mathrm{m^6\,K\,J^{-1}\,mol^{-2}}, \tag{S12}$$

where $A\exp(-E_a/RT)$ describes the atomic mobility and $\exp(-\Delta G^*/k_B T)$ is the equilibrium fraction of sites occupied by critical nuclei. Because the

barrier scales as $\gamma^3/\Delta\mu^2$, it vanishes in vacuum, where the water pressure approaches zero and $\Delta\mu$ grows without bound, but is significant at water saturation, where $\Delta\mu$ is only 6–11 kJ (Table S4).

Internal nucleation. Goethite dehydroxylates topotactically: hematite nucleates within the parent crystal, inherits its oxygen framework and orientation, and the pseudomorph develops pores and twinned hematite domains (Lima-de Faria, 1963; Watari et al., 1979, 1983; Cudennec and Lecerf, 2005). We therefore consider only internal nucleation, in which the new interface of the burgeoning nuclei is between hematite and goethite. We know of no measured or calculated energy for that interface, so γ is replaced by an effective parameter γ_{eff}, calibrated from the data.

Calibration. When Farley et al. (2024) isolated the liquid nitrogen trap from the heated goethite, they raised $P_{\text{H}_2\text{O}}$ in the line and lowered $\Delta\mu$, and the isothermal rate fell by a factor s (§4.1). With A and E_a unchanged, Eq. (S12) gives

$$\gamma_{\text{eff}} = \left[\frac{T\,\Delta\mu_{\text{off}}^2\ln(1/s)}{C}\right]^{1/3}, \tag{S13}$$

where $\Delta\mu_{\text{off}}$ is evaluated at an upper bound on the water pressure with the trap isolated. This pressure estimate of the water released depends on the volume of the experimental apparatus and the water content emitted by the goethite, which was measured with capacitance manometry (§S5.1). With the trap in place, the water pressure approaches zero, the driving force is unbounded, and its contribution to the barrier vanishes. Any finite water pressure in the trapped line would give a larger γ_{eff}, so this limit is conservative. This gives $\gamma_{\text{eff}} = 0.14$ J m^{-2} (range 0.12–0.15). In vacuum the barrier vanishes, so conversion is barrierless, consistent with the simple Arrhenius behavior of both vacuum experiments. At saturation and 250 °C, the critical nucleus has a radius of 0.7–0.9 nm (26–52 formula units) and the barrier is 32–63 k_BT. For nuclei this small, classical nucleation theory has no literal meaning, so γ_{eff} is best read as the single parameter of a rate law in which $\ln k$ falls as $-1/\Delta\mu^2$, not as an interface energy.

Translation to water saturation. For the same sample, only $\Delta\mu$ changes between vacuum and saturation:

$$\frac{k_{\text{sat}}}{k_{\text{vac}}} = \exp\left[-\frac{C\gamma_{\text{eff}}^3}{T}\frac{1}{\Delta\mu_{\text{sat}}^2}\right]. \tag{S14}$$

Applied step by step to the water vapor saturated schedule, Eq. (S14) predicts a total conversion ($\sum_i \frac{k_{\text{sat}}}{k_{\text{vac}}}(T_i)\,k_{\text{vac}}(T_i)\,\Delta t_i$) of $10^{-12.4}$, $10^{-20.7}$ and $10^{-25.8}$

for $\gamma_{\text{eff}} = 0.12$, 0.14 and 0.15 $\mathrm{J\,m^{-2}}$, and $10^{-38.7}$ to $10^{-9.1}$ (95%) when the uncertainties in the calibration and in the thermodynamic data are propagated (§S5.1). The exact values reflect the assumed $\Delta\mu^{-2}$ dependence; what matters is that the model predicts suppression far beyond the $10^{-3.8}$ that the data require (main text, §4.3).

Assumptions. This is a consistent physical explanation of the data, not a proof. It is not exhaustive. This analysis assumes (i) that the trap-isolated slowdown acts through $\Delta\mu$, rather than, for example, through water blocking reactive sites; (ii) that γ_{eff}, A and E_a are the same in vacuum and in water; (iii) that the barrier depends on the bulk driving force without significant contribution from surface terms (this analysis extended to 10 nm grains raises the predicted conversion only to $\sim 10^{-13}$; §S5.1); and (iv) that conversion is solid-state. Conversion by dissolution of goethite and precipitation of hematite from the fluid is not described by Eq. (S14) (see §S7). The calibrated γ_{eff} absorbs contributions from the sample's microstructure and cannot be transferred to other goethites.

S5.1. Calibration of γ_{eff}

Equation (S13) follows from Eq. (S12) applied to a trap-isolated step and its trapped neighbors, $\ln(1/s) = C\gamma_{\text{eff}}^3/(T\,\Delta\mu_{\text{off}}^2)$; the corresponding term for the trapped neighbors vanishes because their driving force is unbounded (item 2). The driving forces are evaluated as follows.

1. $\Delta\mu_{\text{off}}$ is a bulk driving force, $-\Delta G_r$ from Eq. (S4) (§S4) with $\Delta G_{\text{surf}} = 0$, at the step temperature and with the solids at vacuum.
2. With the trap in place, the water pressure in the line approaches zero, so the chemical potential of water falls without bound, $\Delta\mu_{\text{on}}$ is unbounded, and its term $\Delta\mu_{\text{on}}^{-2}$ vanishes. This is the conservative limit: any finite water pressure in the trapped line would give a larger γ_{eff} and a stronger predicted slowdown at saturation. $\Delta\mu_{\text{off}}$ is evaluated at the water pressure reached with the trap isolated (below).
3. Surface terms are omitted. The barrier of an internal nucleus depends on the free energy per volume of material transformed, not on the external surface of the host grain. The external surface energy of the grain enters the overall thermodynamic stability (§S4), but is not a part of the nucleation barrier, which is evaluated for internal nuclei.

We bound the water pressure with the trap isolated by assuming that all evolved water was in the vapor, distributed uniformly through an apparatus

of volume V_{app} at 298 K:

$$P_{\mathrm{H_2O}} = \frac{(m_{\mathrm{w}}/M_{\mathrm{w}})\,RT}{V_{\mathrm{app}}}. \tag{S15}$$

With $m_{\mathrm{w}} = 0.027$ mg at 230 °C and 0.016 mg at 240 °C (1.50 and 0.89 μmol), and $190 \le V_{\mathrm{app}} \le 600$ mL, Table S5 gives the pressures and the resulting γ_{eff}. The pressures are end-of-step values; we also evaluate half of each as a time average for steady accumulation. This is an upper bound on the pressure, because some water adsorbs on the walls of the line; a lower true pressure gives a larger γ_{eff} and more suppression at saturation, so the bound is conservative. Pressure enters only through $\ln P$ in $\Delta\mu_{\mathrm{off}}$ and then through $\Delta\mu_{\mathrm{off}}^{2/3}$, so a threefold uncertainty in volume changes γ_{eff} by only about 5%. The two steps agree within 20%, and we adopt $\gamma_{\mathrm{eff}} = 0.14$ J m^{-2}, with a range of 0.12–0.15 J m^{-2}.

Table S5: Water pressure with the trap isolated and the calibrated γ_{eff} (J m^{-2}; Eq. (S13)).

T (°C)	s	V_{app} (mL)	$P_{\mathrm{H_2O}}$ (Pa)	γ_{eff}, end-of-step P	γ_{eff}, $P/2$
230	0.333	190	19.6	0.120	0.124
230	0.333	600	6.2	0.126	0.130
240	0.224	190	11.6	0.140	0.145
240	0.224	600	3.7	0.147	0.152

S5.2. Uncertainty propagation

The Monte Carlo range quoted in §4.3 (4000 draws) samples the trap-isolated step (230 or 240 °C); s, log-uniformly between 0.30–0.37 at 230 °C and 0.22–0.23 at 240 °C; the apparatus volume, log-uniformly between 190 and 600 mL; a time-averaging coefficient for the pressure between 0.5 and 1 to account for how quickly the evolved water accumulates in the diffusion cell, and a common offset of ΔG_r, uniformly within ±2.8 kJ per reaction (the ±1.4 kJ per mole FeOOH of Majzlan et al. 2003a), which shifts every driving force together. The offset barely changes γ_{eff} (95% range 0.12–0.15 J m^{-2}) but contributes most of the spread in the predicted conversion, because it enters the small $\Delta\mu_{\mathrm{sat}}$ (6–11 kJ) squared.

S6. Multiple Diffusion Domain Fits and Thermal History Results

The modeling procedure is described in the main text (§2.2) and follows Farley and Flowers (2012) and Shuster and Farley (2004).

S6.1. Multiple diffusion domain fits for $N = 1$–5

Figure S6 shows the fitted fractional and cumulative ^{3}He yields, and the Arrhenius and delta plots they imply, for one to five domains. The fits for all N are summarized in Table S6.

Table S6: Multiple diffusion domain fits to the ^{3}He release of HT-Win-06-01b (N domains, $k = 2N - 1$ parameters). RMS is the yield misfit ($\times 10^{-4}$, fraction of total ^{3}He) over the 28 steps above the signal threshold. "Fast" is the ^{3}He fraction in domains with $r/r_0 < 0.1$; f_{main} and r_{main}/r_0 are the fraction and size of the largest-fraction domain. For $N \geq 5$, the dominant domain is split among several domains of identical size.

N	k	RMS	Arrh. χ^2_ν	fast (%)	f_{main} (%)	r_{main}/r_0
1	1	85.30	27.45	0.0	100.0	0.80
2	3	20.20	3.30	2.2	97.8	1.05
3	5	8.45	1.08	2.7	97.3	1.12
4	7	7.81	1.00	2.8	97.2	1.12
5	9	7.81	1.11	2.8	95.1	1.12
6	11	7.81	1.24	2.8	85.6	1.12
7	13	7.81	1.40	2.8	85.3	1.12
8	15	7.81	1.62	2.8	76.8	1.12
9	17	7.81	1.91	2.8	76.5	1.12
10	19	7.81	2.34	2.8	74.6	1.12

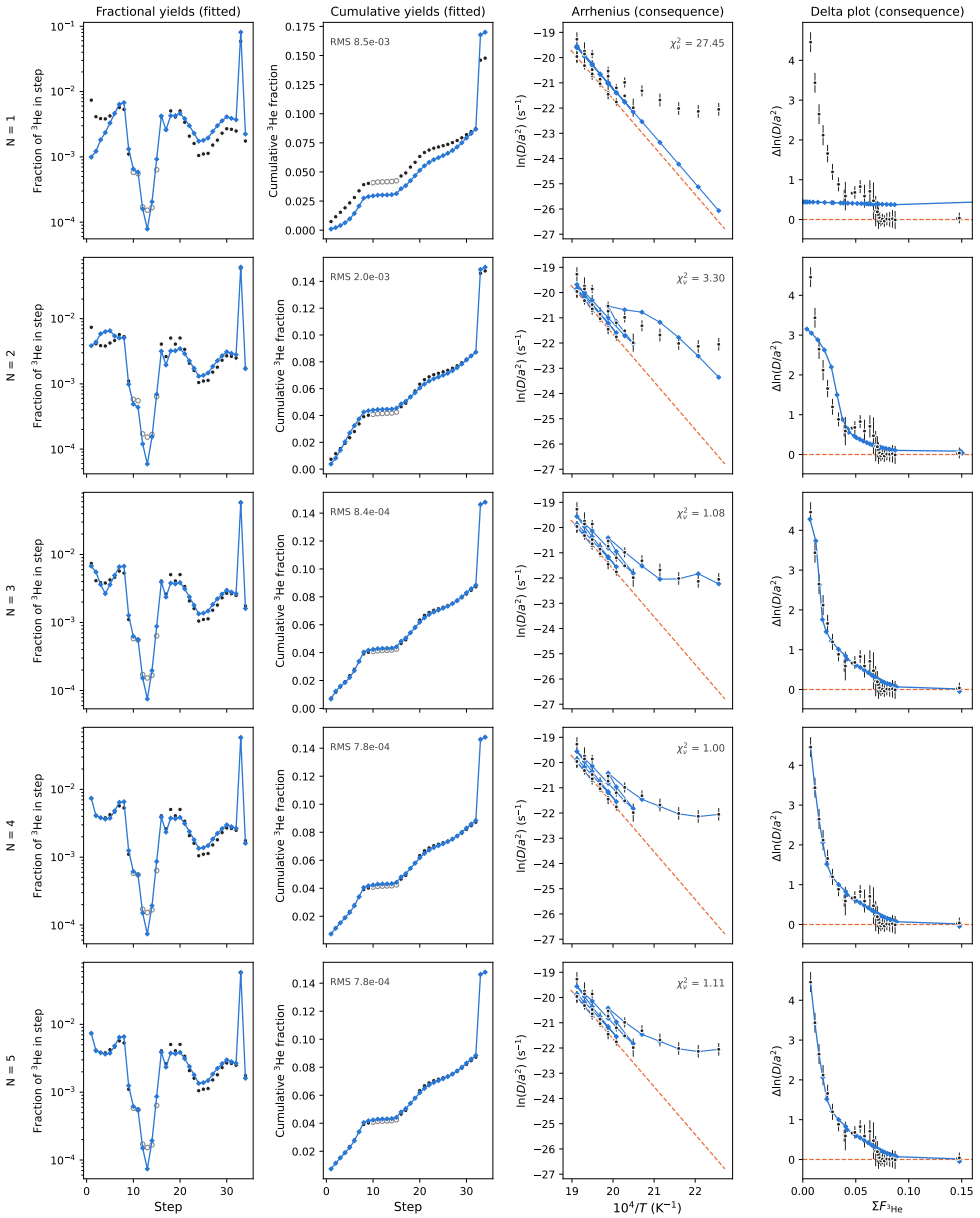


Figure S6: Multiple diffusion domain fits to the ^{3}He release of HT-Win-06-01b for $N = 1$–5 domains (rows). Columns: fitted fractional and cumulative yields, and the Arrhenius and delta plots implied by each fit. Hollow grey symbols are the six low-signal steps, which are not fitted and are omitted from the Arrhenius and delta plots.

Table S7: Thermal histories fitted to the ^{4}He/^{3}He spectrum of HT-Win-06-01b with the four-domain structure fixed. Each row gives the best path with a predicted He age of 130–180 Ma along with the reduced χ^2 value ($\nu = 22$ for steady cooling and the isothermal hold, $\nu = 21$ for burial). For steady cooling, $T_{\max} = T_0$ is reached at crystallization, so $t_{\text{peak}} = t_{\text{form}}$. For the isothermal hold, $T_{\max} = T_{\text{iso}}$ throughout. Burial crystallization ages are fixed within each row. The last three columns give the ranges of the age-consistent paths within the joint 1σ region about the best fit (main text, §2.2).

Path	t_{form} (Ma)	t_{peak} (Ma)	$T_{\max}$ (°C)	χ^2_ν	t_{He} (Ma)	t_{form} (1σ)	t_{peak} (1σ)	$T_{\max}$ (1σ)
Linear cooling	150	150	78	2.00	132	150–175	150–175	76–80
Burial	180	70	61	5.36	164	fixed	60–82	58–65
Burial	195	80	61	5.04	177	fixed	68–98	58–65
Burial	210	115	71	6.53	180	fixed	105–130	70–74
Isothermal hold	150	–	41	18.5	141	140–190	–	39–43

S7. Fluid-Mediated Conversion of Synthetic Goethite Pigment

S7.1. Methods

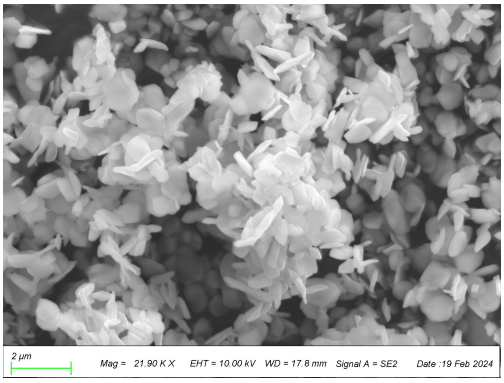
Synthetic goethite pigment, a commercial yellow iron-oxide pigment of acicular crystals ∼1 µm long, was treated in the same way as the Bahia goethite (main text, §2.3) but only with water, without vacuum or open-air runs: at least five larger silica glass ampoules, each loaded with pigment and an excess of water and fused shut at ambient pressure, were held at 200–250 °C for about a day in an oven whose voltage was adjusted by hand. These runs were a rapid, qualitative test; details of individual runs (temperature histories, durations, water-to-solid ratios) were not recorded, and reaction was assessed by color (yellow goethite, red hematite) before imaging.

S7.2. Results and Interpretation

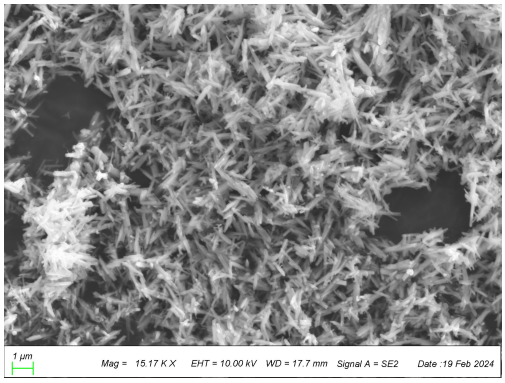
We observed the spontaneous conversion of synthetic goethite pigment (§S7.1) to hematite when heated in ampoules with excess water at experimental temperatures. Roughly equal proportions of ampoules were completely reacted (red), completely unreacted (yellow), or contained both reacted and unreacted material in isolated patches. In the mixed ampoules, the red and yellow portions were entirely isolated from one another, separated by gas bubbles. Unreacted pigment retains its original acicular habit (Fig. S7b). The reacted hematite product is instead composed of equant to platy crystals ∼0.5–1 µm across (Fig. S7a). The change of shape rules out pseudomorphic dehydroxylation like that observed to occur in goethite heated in vacuum (Fig. 1), indicating dissolution of goethite and nucleation and growth of hematite from the fluid, a pathway outside the solid-state nucleation model developed in §4.3.

Conversion either went to completion within a fluid-connected volume or did not occur at all, even where reacted and unreacted pigment shared the same ampoule and thermal history. We posit that the fluid-mediated reaction requires some threshold of dissolved Fe(III) in the fluid to nucleate hematite, but because the reaction consumes and produces dissolved Fe(III) in equal proportions, it is not self-limiting and proceeds quickly to completion once that threshold is reached.

Grain size and bulk thermodynamics cannot explain why the pigment blades reacted when the natural polycrystalline goethite did not. At 1 µm, the surface contribution to ΔG_r is only ∼0.04 kJ, and the bulk driving force at saturation is the same 6–11 kJ for both materials (Table S4). The most



(a) Reacted (red) product



(b) Unreacted (yellow) pigment

Figure S7: Secondary electron images of synthetic goethite pigment after treatment at water vapor saturation and 200–250 °C for about a day (§S7.1). (a) Material from a completely reacted (red) ampoule: the acicular goethite has been replaced by equant to platy hematite crystals ∼0.5–1 µm across. (b) Material from an unreacted (yellow) ampoule, which retains the unchanged acicular habit of the starting pigment. The change in crystal habit requires dissolution of goethite and precipitation of hematite from the fluid, and the all-or-nothing outcome of the experiments indicates that the reaction is fast once it begins.

likely difference is the fluid itself. Commercial iron-oxide pigments are commonly produced by precipitation from iron-salt solutions. They can carry residual soluble salts such as iron and alkali sulfates or nitrates, and they are often surface-treated with organometallic compounds (Legodi and de Waal, 2007; de Almeida Silva et al., 2017; Xu et al., 2021). Such contaminants would change the pH and ionic strength of the water they are heated in, and with them the solubility of Fe(III). The dissolution–precipitation pathway is limited by how much iron the fluid carries, so a modest increase in Fe(III) solubility could make a nucleation-limited reaction viable. The natural Win-06-01b goethite, by contrast, was heated in distilled water, where near-neutral Fe(III) solubility is very low, potentially offering an explanation for the divergent outcomes.

References

de Almeida Silva, R., Menezes, J.C.S.d.S., Lopes, F.A., Kirchheim, A.P., Schneider, I.A.H., 2017. Synthesis of a goethite pigment by selective precipitation of iron from acidic coal mine drainage. Mine Water and the

Environment 36, 386–392. URL: https://doi.org/10.1007/s10230-017-0448-2, doi:10.1007/s10230-017-0448-2.

Becker, R., Döring, W., 1935. Kinetische Behandlung der Keimbildung in übersättigten Dämpfen. Annalen der Physik 416, 719–752.

Bell, I.H., Wronski, J., Quoilin, S., Lemort, V., 2014. Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp. Industrial & Engineering Chemistry Research 53, 2498–2508. doi:10.1021/ie4033999.

Berman, R.G., 1988. Internally-consistent thermodynamic data for minerals in the system Na2O-K2O-CaO-MgO-FeO-Fe2O3-Al2O3-SiO2-TiO2-H2O-CO2. Journal of Petrology 29, 445–522. URL: https://doi.org/10.1093/petrology/29.2.445, doi:10.1093/petrology/29.2.445.

Berman, R.G., Brown, T.H., 1985. Heat capacity of minerals in the system Na2O-K2O-CaO-MgO-FeO-Fe2O3-Al2O3-SiO2-TiO2-H2O-CO2: representation, estimation, and high temperature extrapolation. Contributions to Mineralogy and Petrology 89, 168–183.

Christian, J.W., 2002. The Theory of Transformations in Metals and Alloys. 3 ed., Pergamon, Oxford.

Cudennec, Y., Lecerf, A., 2005. Topotactic transformations of goethite and lepidocrocite into hematite and maghemite. Solid State Sciences 7, 520–529.

De Yoreo, J.J., Vekilov, P.G., 2003. Principles of crystal nucleation and growth. Reviews in Mineralogy and Geochemistry 54, 57–93.

Lima-de Faria, J., 1963. Dehydration of goethite and diaspore. Zeitschrift für Kristallographie 119, 176–203. doi:10.1524/zkri.1963.119.3-4.176.

Farley, K.A., Flowers, R.M., 2012. (U–Th)/Ne and multidomain (U–Th)/He systematics of a hydrothermal hematite from eastern Grand Canyon. Earth and Planetary Science Letters 359-360, 131–140.

Farley, K.A., Monteiro, H.B., Vasconcelos, P.M., Waltenberg, K., 2024. Dehydration of goethite during vacuum step-heating and implications for he retentivity characterization. Chemical Geology 663, 122254.

Gibbs, J.W., 1878. On the equilibrium of heterogeneous substances. Transactions of the Connecticut Academy of Arts and Sciences 3, 108–248, 343–524.

Jones, W.M., 1953. Permeability and solubility of He^3 and He^4 in vitreous silica. Journal of the American Chemical Society 75, 3093–3096.

Kashchiev, D., 2000. Nucleation: Basic Theory with Applications. Butterworth-Heinemann, Oxford.

Lasaga, A.C., 1998. Kinetic Theory in the Earth Sciences. Princeton University Press, Princeton.

Legodi, M.A., de Waal, D., 2007. The preparation of magnetite, goethite, hematite and maghemite of pigment quality from mill scale iron waste. Dyes and Pigments 74, 161–168. URL: https://www.sciencedirect.com/science/article/pii/S0143720806000623, doi:10.1016/j.dyepig.2006.01.038.

Majzlan, J., Grevel, K.D., Navrotsky, A., 2003a. Thermodynamics of Fe oxides: Part II. enthalpies of formation and relative stability of goethite (α-FeOOH), lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3). American Mineralogist 88, 855–859.

Majzlan, J., Lang, B.E., Stevens, R., Navrotsky, A., Woodfield, B.F., Boerio-Goates, J., 2003b. Thermodynamics of Fe oxides: Part I. entropy at standard temperature and pressure and heat capacity of goethite (α-FeOOH), lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3). American Mineralogist 88, 846–854.

Masaryk, J.S., 1968. Permeation in fused silica. As cited in the fused-silica helium permeability literature (e.g., Shelby, 1972); original venue not confirmed.

Monteiro, H., 2017. Paleoenvironmental Evolution of Continental Landscapes through Combined High-Resolution Geochronology and $\delta^{18}O$ Ion Microprobe Analysis of Goethite. Ph.D. thesis. The University of Queensland. doi:10.14264/uql.2018.6.

Navrotsky, A., Mazeina, L., Majzlan, J., 2008. Size-driven structural and thermodynamic complexity in iron oxides. Science 319, 1635–1638.

Shelby, J.E., 1972. Helium migration in natural and synthetic vitreous silica. Journal of the American Ceramic Society 55, 61–64.

Shuster, D.L., Farley, K.A., 2004. 4he/3he thermochronometry. Earth and Planetary Science Letters 217, 1–17. doi:10.1016/S0012-821X(03)00595-8.

Swets, D.E., Lee, R.W., Frank, R.C., 1961. Diffusion coefficients of helium in fused quartz. The Journal of Chemical Physics 34, 17–22.

Thyer, B.J., 2026. Recipes for Rare Metal Ores: Magmas, Metasomatism, and Meteoric Mobilization via Sulfide/Goethite (U-Th)/He Geochronology and ML-Accelerated Thermodynamic Modeling. Ph.D. thesis. California Institute of Technology. Pasadena, CA, USA.

Thyer, B.J., Treffkorn, J., Farley, K.A., 2026. A simple method for helium step heat experiments at high water or sulfur fugacity. Geochronology Discussions In preparation.

Turnbull, D., Fisher, J.C., 1949. Rate of nucleation in condensed systems. The Journal of Chemical Physics 17, 71–73. URL: https://pubs.aip.org/jcp/article/17/1/71/200648/Rate-of-Nucleation-in-Condensed-Systems, doi:10.1063/1.1747055.

Virtanen, P., et al., 2020. SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods 17, 261–272.

Voigt, D.E., Will, G., 1981. The system fe2o3-h2o under high-pressures. Neues Jahrbuch für Mineralogie, Monatshefte 2, 89.

Volmer, M., Weber, A., 1926. Keimbildung in übersättigten Gebilden. Zeitschrift für Physikalische Chemie 119, 277–301.

Vyazovkin, S., Burnham, A.K., Criado, J.M., Pérez-Maqueda, L.A., Popescu, C., Sbirrazzuoli, N., 2011. ICTAC kinetics committee recommendations for performing kinetic computations on thermal analysis data. Thermochimica Acta 520, 1–19. doi:10.1016/j.tca.2011.03.034.

Wagner, W., Pruß, A., 2002. The IAPWS formulation 1995 for the thermodynamic properties of ordinary water substance for general and scientific use. Journal of Physical and Chemical Reference Data 31, 387–535. doi:10.1063/1.1461829.

Watari, F., Delavignette, P., Van Landuyt, J., Amelinckx, S., 1983. Electron microscopic study of dehydration transformations. III. High resolution observation of the reaction process FeOOH → Fe2O3. Journal of Solid State Chemistry 48, 49–64. doi:10.1016/0022-4596(83)90058-0.

Watari, F., Van Landuyt, J., Delavignette, P., Amelinckx, S., 1979. Electron microscopic study of dehydration transformations. I. Twin formation and mosaic structure in hematite derived from goethite. Journal of Solid State Chemistry 29, 137–150. doi:10.1016/0022-4596(79)90218-4.

Xu, M., Pan, G., Guo, Y., Yang, S., Fang, Z., 2021. Surface modification and structure analysis of coated iron oxide yellow pigments to improve dispersion in organic solvents. Surface and Interface Analysis 53, 933–945. URL: https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/sia.6995, doi:10.1002/sia.6995.